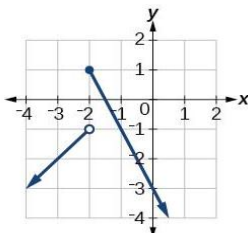
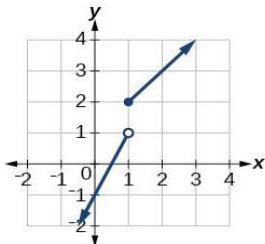


Section 1.2 Exercise Answers

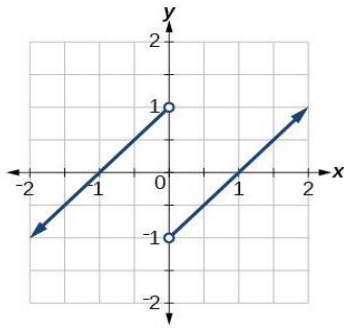
1. $(-\infty, \infty)$
2. $(-\infty, \infty)$
3. $[2, \infty)$
4. $(-\infty, 3]$
5. $(-\infty, \frac{4}{3}]$
6. $(-\infty, \infty)$
7. $(-\infty, \infty)$
8. $(-\infty, \infty)$
9. $(-\infty, 6) \cup (6, \infty)$
10. $(-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$
11. $[-4, 4) \cup (4, \infty)$
12. $(-\infty, -11) \cup (-11, 2) \cup (2, \infty)$
13. $(-\infty, -2) \cup (-2, 3) \cup (3, \infty)$
14. $(-\infty, -3) \cup (-3, 5) \cup (5, \infty)$
15. $(3, \infty)$
16. $(-\infty, 5)$
17. $(6, \infty)$
18. $[6, \infty)$
19. $(-\infty, 0) \cup (0, \infty)$
20. $(-\infty, -9) \cup (-9, 9) \cup (9, \infty)$
21. The domain is $(2, 8]$ and the range is $[6, 8)$.
22. The domain is $[4, 8)$ and the range is $(2, 8]$.
23. The domain is $[-4, 4]$ and the range is $[0, 2]$.
24. The domain is $[2, 6]$ and the range is $[-6, -2]$.
25. The domain is $[-5, 3)$ and the range is $[0, 2]$.
26. The domain is $[-3, 2)$ and the range is $[-5, 4]$.
27. The domain is $(-\infty, 1]$ and the range is $[0, \infty)$.
28. The domain is $[-4, \infty)$ and the range is $[-2, \infty)$.
29. The domain is $(-2.5, \infty)$ and the range is $[-4, \infty)$.
30. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



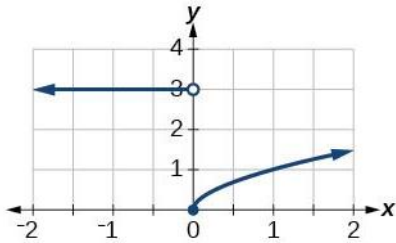
31. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



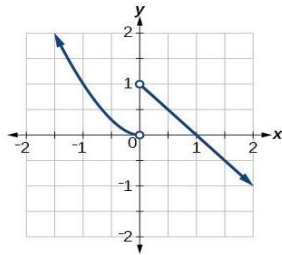
32. The function is defined for all values of x except $x = 0$, so the domain is $(-\infty, 0) \cup (0, \infty)$.



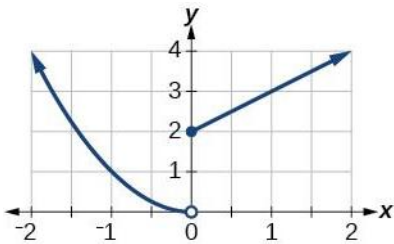
33. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



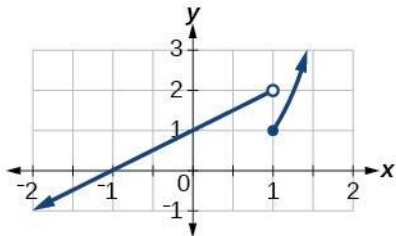
34. The function is defined for all values of x except $x = 0$, so the domain is $(-\infty, 0) \cup (0, \infty)$.



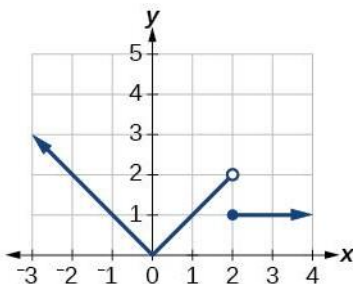
35. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



36. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



37. The function is defined for all values of x , so the domain is $(-\infty, \infty)$.



38. $f(-3) = -2, f(-2) = 1, f(-1) = -1, f(0) = -3$
39. $f(-3) = 1, f(-2) = 0, f(-1) = 0, f(0) = 0$
40. $f(-3) = -15, f(-2) = -5, f(-1) = 1, f(0) = -7$
41. $f(-1) = -4, f(0) = 6, f(2) = 20, f(4) = 34$
42. $f(-1) = -1, f(0) = -2, f(2) = 7, f(4) = 5$
43. $f(-1) = -5, f(0) = 3, f(2) = 3, f(4) = 16$
44. $(-\infty, \infty)$
45. $(-\infty, 1) \cup (1, \infty)$
46. $(-\infty, 0) \cup [2, \infty)$